

GelSlim 5.0: Unifying Vision and Touch with a Single Fingertip Camera

Abstract—Vision and touch provide complementary cues for object pose and local contact geometry, but external cameras are often occluded at the contact site. We present GelSlim 5.0, a fingertip sensor that acquires visual and tactile observations with one fixed camera and a continuous UV-responsive contact layer. By switching illumination, the camera sequentially acquires through-membrane visual images and contact-sensitive tactile images from a shared viewpoint, avoiding inter-camera extrinsic calibration. To combine these observations, we introduce VisTacFusion, a multimodal transformer with separate prediction paths for planar object pose and contact geometry. Asymmetric bottleneck fusion incorporates visual context while preserving tactile spatial features for depth and surface-normal estimation. Modality dropout enables inference with paired inputs or either modality alone. On real validation data from 12 known objects, paired inputs reduce depth MSE, angular error, and translation L_1 by 61.2%, 72.3%, and 61.3%, respectively, relative to tactile-only inference with the same model. Relative to visual-only inference, the corresponding reductions are 93.7%, 6.1%, and 8.6%.

I. INTRODUCTION

Contact-rich manipulation requires estimating both object pose and the local geometry of contact. Vision supplies object context, while optical tactile sensors such as GelSight and GelSlim capture detailed contact structure [1], [2]. The modalities can resolve different ambiguities: visual context can distinguish object poses that produce similar contact patches, while tactile measurements reveal local surface geometry that visual appearance alone may not resolve.

With external cameras, however, the hand and object can occlude the contact site, and viewpoint differences complicate spatial association with tactile measurements. A shared fingertip camera places visual sensing at the contact interface and provides a common frame for relating object appearance to contact geometry. Prior visuotactile sensors obtain scene access through semitransparent skins [3], controllable films [4], UV-visible markers [5], or spatial multiplexing [6]. Building on these designs, we couple a continuous contact interface with an estimator for object pose and spatially resolved contact geometry.

We introduce GelSlim 5.0, a fingertip sensor that co-locates vision and touch within the GelSlim 4.0 body design [7]. A continuous UV-responsive contact layer allows one fixed camera to record through-membrane visual images with UV off and contact-sensitive tactile images with UV on (Fig. 1). The observations are acquired sequentially under approximately stationary contact and share a viewpoint and image coordinates, supporting spatial association without inter-camera extrinsic calibration. Table I positions the sensor among prior interfaces.

To exploit these complementary observations, VisTacFusion uses separate prediction paths for planar object pose and

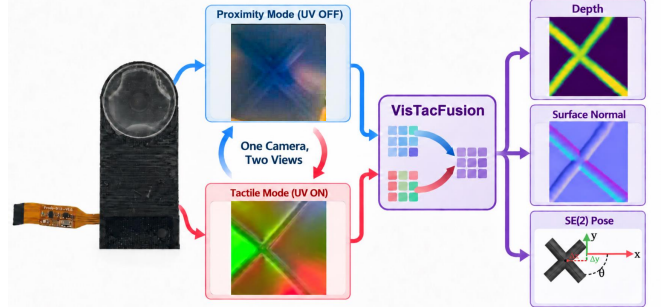


Fig. 1: GelSlim 5.0 enables visual and tactile feedback with a single fingertip camera. By switching illumination, the sensor operates in proximity mode with UV off, allowing the camera to see through the membrane, and in contact-sensitive tactile mode with UV on. Both modes share the same viewpoint and require no inter-camera extrinsic calibration. VisTacFusion combines their complementary signals to estimate planar object rotation and translation, as well as contact depth and surface normals. The observations are acquired sequentially under approximately stationary contact.

contact geometry. Tactile-initialized queries provide a pose estimate, while intermediate tactile features preserve spatial detail for contact-depth and surface-normal estimation. A shared visual bottleneck conditions both paths, allowing object context to inform pose and geometry predictions. Modality dropout trains the model to operate with either input alone or the pair.

Our contributions are (1) a fingertip sensor that co-locates vision and touch using a continuous UV-responsive contact interface; (2) an estimator that combines visual context with separate pose queries and spatial geometry features; and (3) an evaluation on 12 known objects covering modality complementarity, encoder and fusion choices, and training configuration. Within-session validation shows lower contact-depth and pose errors for paired observations than for either input alone in the same estimator.

II. RELATED WORK

From tactile fingers to switchable visuotactile sensing. GelSight, DIGIT, and the GelSlim family image a deformable contact interface [1], [2], [7], [15], [16], [18]. GelSlim 4.0 emphasizes reproducible hardware and perception [7]; GelSlim 5.0 retains its body design and changes the contact film and illumination to add a visual observation state. Scene access has also been obtained with semitransparent interfaces in STS, VisTac, and StereoTac [3], [11], [12]; UV-visible markers in SpecTac and TransTac [5], [13]; PDLC control in ProTac [4]; and mechanical switching in Look-to-Touch [14]. ViTacTip and M³Tac use transparent or spectrally selective constructions [8], [10]. MuxGel spatially interleaves transparent and tactile regions for single-camera

Sensor construction and physical size				Mode selection		Observation and reconstruction		
Sensor	Cameras	Contact interface	Size [mm] H: head; I: interface	Selected element	Control	Tactile cue	Contact representation	Acquisition
<i>Persistent transparency or parallel spectral channels</i>								
ViTacTip [8], [9]	1 shared	Clear skin; marker-tipped pins	H: NR I: 2500 mm^2 ^S	None	None	Pin / marker motion	Marker deformation	Same frame
M ³ Tac [10]	3 bands ^a	Spectrally selective membrane	H: $\varnothing 65 \times 80$ ^R I: film $t = 0.2$ ^R ; area NR	Spectral channels	Filters + LEDs	Photometric shading	Surface reconstruction	Parallel bands
<i>Fixed interface with illumination-dependent observations</i>								
STS [3]	1 shared	Semitransparent reflective skin	H: NR I: 150×150 ^R pad ($t = 5$)	Lighting ratio	LEDs	Photometric shading	Contact images	Alternating / mixed
StereoTac [11]	2 stereo	Semitransparent reflective skin	H: NR I: square; side NR	Illumination	LEDs	Photometric shading	Surface depth	Alternating
VisTac [12]	2 paths	Semitransparent reflective skin	H: $70 \times 38 \times 33$ ^R I: 45×34 ^R pad; 32×32 ^R active	Illumination	LEDs	Photometric shading	Surface depth	Sequential
SpecTac [5]	1 shared	Clear gel; fluorescent markers	H: NR I: $\varnothing$ NR; $t \approx 10 + 0.5$ ^R	Marker visibility	UV LEDs	Marker motion	Marker deformation	UV-selected
TransTac [13]	2 stereo	Clear gel; UV markers	H: NR I: $t = 5$ ^R ; lateral NR	Marker visibility	UV LEDs	Stereo marker motion	3D marker geometry	UV-selected
<i>Auxiliary transparency control or mechanical interface selection</i>								
ProTac [4]	2 views	Clear skin; PDLC film; markers	H: NR; I: NR	Auxiliary PDLC	Voltage	Marker motion	Skin deformation	Alternating
Look-to-Touch [14]	1 shared	Clear / opaque belt regions	H: NR; I: NR	Exposed belt region	Servo	Photometric shading	Surface depth	Sequential
<i>No switching – GelSlim family</i>								
GelSlim (1.0) [2]	1 tactile	Opaque reflective skin	H: NR I: 50×50 ^R pad	None	None	Photometric shading	Contact images	Tactile only
GelSlim 2.0 [15], [16]	1 tactile	Opaque skin; markers	H: $50 \times 172 \times 25$ ^S I: 1200 mm^2 ^S	None	None	Shading + marker motion	Marker deformation ^b	Tactile only
GelSlim 3.0 [16]	1 tactile	Opaque skin; markers	H: $37 \times 80 \times 20$ ^R I: 675 mm^2 ^R	None	None	Shading + marker motion	Surface depth	Tactile only
GelSlim 4.0 [7], [17]	1 tactile	Opaque skin; optional markers	Body: $83.6 \times 20.1 \times 35.1$ ^C I: NR	None	None	Photometric shading	Surface depth	Tactile only
GelSlim 5.0 (ours)	1 shared	Continuous UV-responsive film^c	$83.6 \times 22.1 \times 35.1$^{A,C} UV layer; I: NR	Contact-film response	UV LEDs	Contact intensity / shading	Learned depth + normals	Sequential

Size guide. H denotes the reported sensing-head envelope; I denotes the contact/active interface; NR means not reported; t is thickness. Dimensions retain the reported axis order; areas are explicitly in mm^2 . ^RPrimary-paper value; ^Slater scholarly report; ^Cofficial body-CAD bounding box; ^Aauthor-provided design constraint. GelSlim 2.0 dimensions refer to the standard prototype summarized in GelSlim 3.0. GelSlim 4.0 reports the gripper-body CAD. GelSlim 5.0 adds the author-specified UV layer along the 20.1 mm axis, giving a nominal $83.6 \times 22.1 \times 35.1$ mm body-plus-layer envelope. Neither value describes a complete assembly or contact area; separate back covers, cables, and robot adapters are excluded. SpecTac lists gel and protective-layer thickness separately, not their summed thickness. A platform pad, camera baseline, or belt stock is not a head envelope.

Reading guide. Geometry refers to the contact interface, not external-scene depth. Output labels describe cited demonstrations rather than limits on other possible outputs; photometric methods may estimate normals or gradients internally. “UV-selected” does not assert same-exposure separation. ^aVisible, near-infrared, and mid-infrared imagers, not three RGB cameras. ^bGelSlim 2.0 also estimates force distributions using inverse FEM. ^cUV-responsive denotes the observed imaging-state change; transmission, reflection, and UV emission have not been separately quantified. Our reconstruction is evaluated on known-object, within-session validation samples.

TABLE I: **Representative sensor designs and physical sizes for scene and contact perception.** Rows summarize cited implementations, not a common accuracy benchmark.

reconstruction [6], while FingerEye uses binocular visuotactile sensing for manipulation [19]. Our implementation combines a continuous UV-responsive outer contact film, sequential same-camera acquisition, and learned pose/contact estimation within the GelSlim body. The estimator uses contact-sensitive image appearance rather than marker displacement, without spatially partitioning the film into clear and coated windows. Table I compares construction, acquisition, and contact representation; head dimensions and contact-interface measurements are listed separately within the size column.

Learning from visual and tactile representations Visual pretraining methods such as MAE, DINO, Depth Anything, CLIP, and SigLIP provide distinct feature representations [20]–[25]. Tactile and cross-modal representations include Sparsh, T3, SITR, TVL, and Sparsh-X [26]–[30], while MViTac learns visual–tactile alignment by contrastive pretraining [31]. We use these representations for object-pose and contact-geometry estimation, rather than introducing a new backbone-pretraining objective. Building on attention bottlenecks [32] and DPT-style dense decoding [33], VisTacFusion uses compact visual context to condition a pose path and a separate contact-geometry decoder. Simulators

such as TACTO motivate supplementing real tactile data with synthetic observations [34]; our evaluation uses Blender-based co-training, not simulation-only transfer.

III. GELSLIM 5.0: COMMON-VIEW SENSING

GelSlim 5.0 is a monocular visual–tactile fingertip sensor that captures both observation modalities from a single camera by switching UV illumination (Fig. 2). The design has three elements: (i) a compliant contact membrane whose optical state changes under UV, (ii) an illumination ring that drives the switching, and (iii) a 3D-printed body housing one camera and a custom driver PCB. Below, we describe each element and the resulting acquisition protocol.

Compliant contact interface. The prototype retains the GelSlim 4.0 gripper-body CAD [7], [17] and uses a two-layer Solaris (Smooth-On) silicone membrane. We form a 5 mm clear base by mixing the silicone 1:1 by weight, vacuum-degassing, and casting it in a mold. The continuous UV-responsive layer is spin-coated from 3 g Solaris mixed with 0.3 mL invisible UV-reactive ink (United Nuclear). This layer contacts the object directly, without a separate protective overlayer.

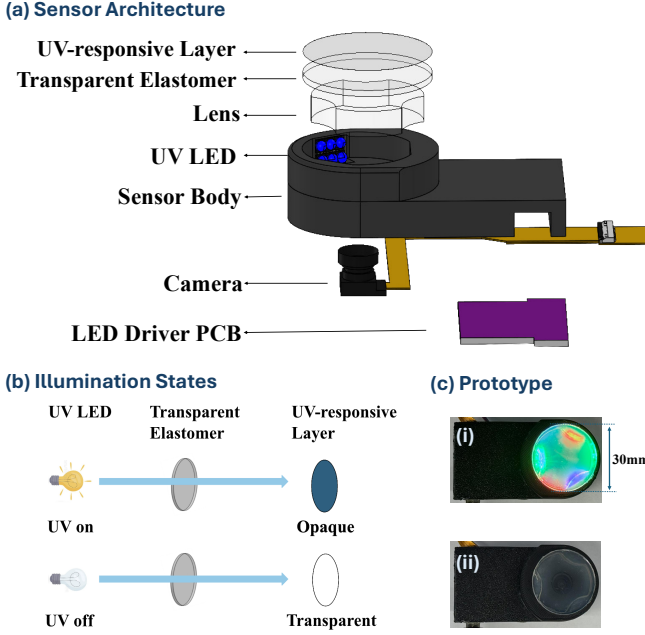


Fig. 2: **GelSlim 5.0 hardware and observation states.** (a) Exploded assembly with the UV-responsive film at the contact interface. (b) Illumination-controlled visual and tactile acquisition. (c) Prototype in (i) tactile mode and (ii) proximity mode.; scale marker: 30 mm.

The 3D-printed PLA body has a nominal circular sensing area of 30 mm diameter and houses a Raspberry Pi camera. An illumination ring uses UV LEDs in place of the blue channel; red and green illumination remain available. A custom driver PCB and a GPIO-controlled MOSFET switch the UV channel and coordinate image capture.

Visual and tactile acquisition With UV off, the camera records a through-membrane visual image I_v ; we refer to this state as proximity mode. With UV on, it records a contact-sensitive tactile image I_t . We use UV-responsive to describe this observed change in imaging state; transmission, reflection, and UV-induced emission have not been separately quantified.

Each pair is captured sequentially under approximately stationary contact. Because the camera and lens remain fixed, the images share a viewpoint and image coordinates, supporting spatial association of appearance and contact features without inter-camera extrinsic calibration (Fig. 4). Refraction, membrane deformation, and residual motion can still affect pixelwise alignment. Camera intrinsics and object-pose calibration remain relevant to estimation.

IV. LEARNING OBJECT POSE AND CONTACT GEOMETRY

VisTacFusion combines visual context with tactile signatures through separate pose and geometry prediction paths (Fig. 3). The pose path aggregates features into a pose vector, while the geometry path preserves spatial detail for depth and normals. Cross-attention associates the modalities without requiring pixelwise correspondence.

A. Problem Formulation

For a known object o , let I_v and I_t denote the image pair acquired under the quasi-static protocol of Sec. III. The input

$\mathcal{I}_m$ is I_v , I_t , or (I_v, I_t) for modes $m \in \{\text{Vis}, \text{Tac}, \text{Both}\}$, respectively. We learn an estimator with parameters ϕ ,

$$f_\phi(\mathcal{I}_m, o) = (\hat{\mathbf{y}}, \hat{D}, \hat{\mathbf{N}}), \quad (1)$$

where $\mathbf{y} = (\cos \theta, \sin \theta, t_x, t_y)$ encodes sensor-relative planar pose. Depth $D \in \mathbb{R}^{H \times W}$ and surface normals $\mathbf{N} \in \mathbb{R}^{H \times W \times 3}$ describe contact geometry on the tactile image grid. Translations are normalized as in Sec. V-A; object identity o is supplied to the pose queries.

B. Modality-Specific Features

Frozen Sitr-B/16 [28] and MAE-L/16 [20] encode touch and vision, respectively. At 224×224 resolution, each produces 196 patch tokens and one CLS token, with widths 768 and 1024. Sitr uses a fixed set of 18 calibration references; calibration tokens are discarded before downstream prediction.

Separate linear projections map both streams to width 768 and add modality embeddings. Tactile patch tokens receive spatial embeddings and initialize 196 spatial queries; the projected CLS token initializes one pose query. All queries receive an object-identity embedding. Projected visual tokens form fixed memory $M \in \mathbb{R}^{197 \times 768}$. All downstream modules are trained jointly.

C. Object Pose Estimation

Tactile-initialized queries $Q_0 \in \mathbb{R}^{197 \times 768}$ read visual context through 16 learned bottleneck tokens $B_0 \in \mathbb{R}^{16 \times 768}$. Define the pre-normalized attention residual

$$\mathcal{R}(X, Y) = X + \text{MHA}(\text{LN}_q(X), \text{LN}_{kv}(Y)), \quad (2)$$

where X supplies queries and Y supplies keys and values, and let $\mathcal{F}(X) = X + \text{FFN}(\text{LN}(X))$. Four layers, indexed by $\ell \in \{1, \dots, 4\}$, update

$$B_\ell = \mathcal{R}_{B, \ell}(B_{\ell-1}, M), \quad (3)$$

$$\tilde{Q}_\ell = \mathcal{F}_{1, \ell}(\mathcal{R}_{Q, \ell}(Q_{\ell-1}, B_\ell)), \quad (4)$$

$$Q_\ell = \mathcal{F}_{2, \ell}(\mathcal{R}_{S, \ell}(\tilde{Q}_\ell, \tilde{Q}_\ell)). \quad (5)$$

Each layer updates the bottleneck from visual memory, then applies query-to-bottleneck cross-attention and query self-attention. States B_ℓ and Q_ℓ carry across layers; only B_0 is a shared learned initialization. Attention uses eight heads. Query FFNs expand the width fourfold with GELU and dropout; the bottleneck update has no FFN.

Writing $Q_4 = [s_{4,1}, \dots, s_{4,196}, p_4]$, the pose head concatenates the final pose token with the mean of the 196 spatial tokens

$$z_{\text{pose}} = [p_4; \text{mean}_j s_{4,j}] \in \mathbb{R}^{1536}. \quad (6)$$

A layer-normalized three-layer MLP maps this vector to two rotation components and two translations. The rotation components are normalized to form $(\cos \hat{\theta}, \sin \hat{\theta})$.

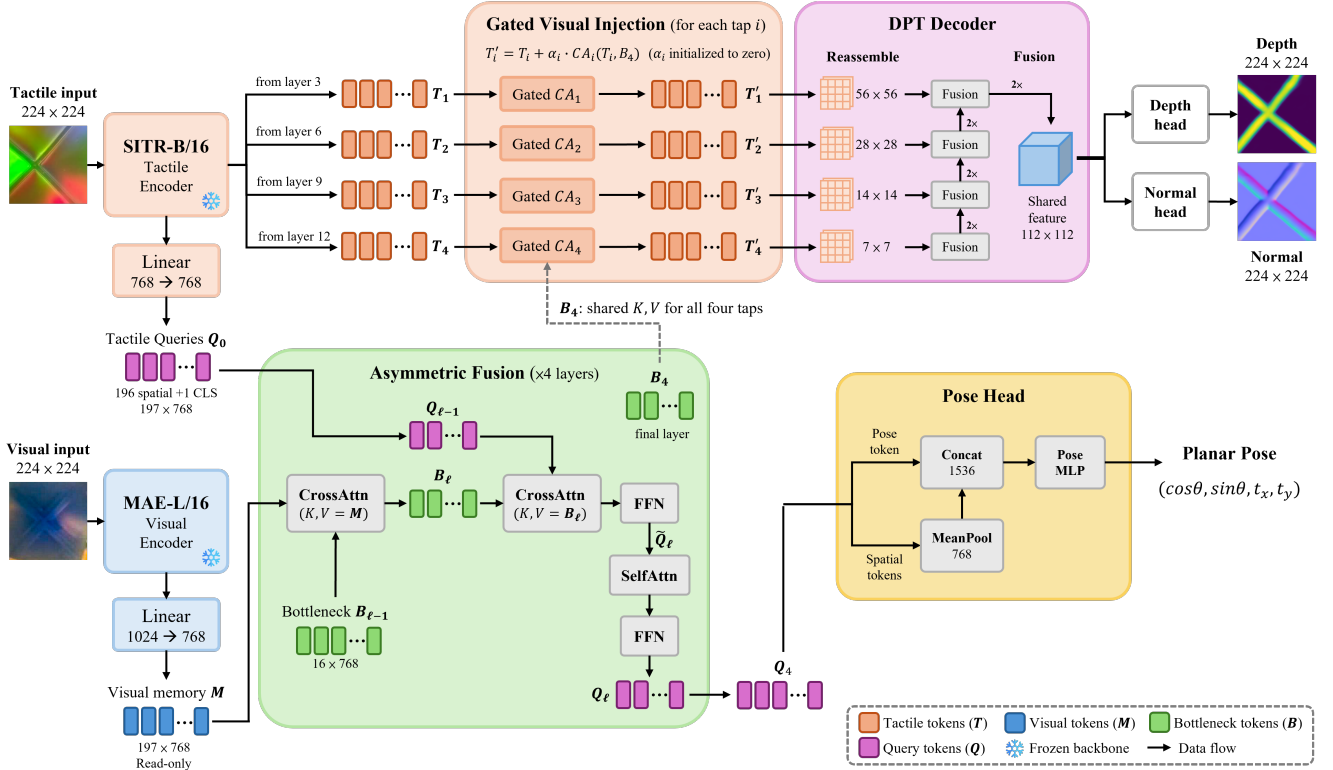


Fig. 3: **VisTacFusion: task-specific estimation of object pose and contact geometry.** Frozen encoders provide tactile features and visual memory M . The pose path uses a 16-token visual bottleneck to condition tactile-initialized queries over four layers; the final query state Q_4 predicts planar pose. The geometry path retains tactile maps from blocks 3, 6, 9, and 12, applies gated conditioning from visual-context state B_4 , and decodes contact depth and surface normals with DPT. Calibration inputs, embeddings, and pre-normalized residual connections are omitted. Geometry thumbnails illustrate estimation targets; missing-input routing is defined in Sec. IV-E.

D. Contact Geometry Estimation

To preserve local contact structure, the geometry path uses intermediate tactile features directly. Four feature maps from encoder blocks 3, 6, 9, and 12 bypass the pose-query trunk. After adding position embeddings, each map $T_i \in \mathbb{R}^{196 \times 768}$ receives context from the final visual bottleneck B_4 :

$$T'_i = T_i + \alpha_i \text{CA}_i(T_i, B_4), \quad i = 1, \dots, 4. \quad (7)$$

Each map has a separate pre-normalized cross-attention module CA_i , with T_i as queries and B_4 as key-values. The learned scalar gates α_i are initialized as zero, so visual conditioning is learned as a residual update to the tactile features. Both pose and geometry losses supervise the shared visual summary B_4 . The geometry decoder uses the conditioned spatial features directly, without depending on pose predictions.

A DPT-style decoder [33] reassembles the four conditioned maps into 256-channel maps at 56^2 , 28^2 , 14^2 , and 7^2 resolutions. Coarse-to-fine fusion produces a shared 112×112 feature map. Separate convolutional heads predict depth and surface normals at 224×224 ; predicted normal vectors are not explicitly normalized.

E. Input Availability and Joint Training

Modality dropout trains the same model with paired or individual inputs. Both follows the architecture above. In *Vis*, learned mask queries replace tactile queries, and MAE

maps from blocks 3, 6, 9, and 12 (projected from 1024 to 768 channels) supply the geometry features. In *Tac*, both fusion cross-attention updates and the geometry path’s visual conditioning are skipped.

All three modes receive pose and geometry supervision. For inputs that include vision, geometry-path visual conditioning is independently enabled with probability p_{inject} during training and always enabled at inference.

Depth and normal losses are full-image MSEs. Rotation uses $1 - \cos(\hat{\theta} - \theta)$, and translation uses coordinate-averaged L_1 error in normalized units. For $k \in \{D, N, \text{rot}, \text{trans}\}$, the uncertainty-weighted training objective [35] is

$$\mathcal{L} = \sum_k (e^{-s_k} \mathcal{L}_k + \frac{1}{2} s_k), \quad (8)$$

where each learned scalar s_k weights one task.

V. EXPERIMENTAL EVALUATION

We evaluate modality complementarity, encoder selection, fusion design, and training-data composition under a shared real-data validation protocol.

A. Sensor Observations and Evaluation Protocol

Sensor observations and endurance Figure 4 illustrates common-view visual and tactile observations of four everyday objects. In the reported cyclic-loading test, a spherical indenter applied more than 45,000 normal indentations at 20 N without observed cracking, delamination, or loss of

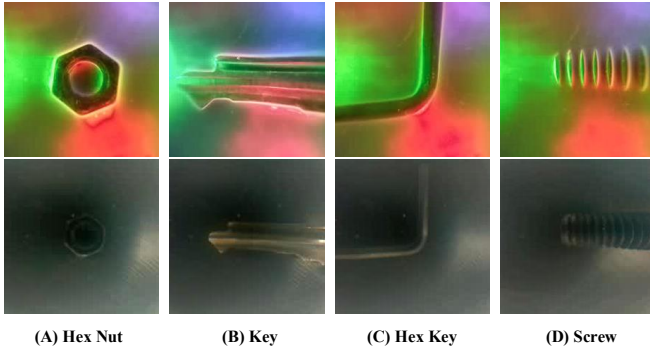


Fig. 4: **Common-view visual-tactile observations.** UV-on tactile images (top) and UV-off proximity images (bottom) are acquired sequentially from the same fixed camera under quasi-static contact.

Model	Geometry		Pose		
	Depth	Normal	Rot.	Angle	Trans.
	MSE ↓	MSE ↓	L_1 ↓	(°) ↓	L_1 ↓
<i>(a) Vis only</i>					
MAE [20]	0.0756	0.0403	0.0135	0.678	0.0060
DINOv3 [22]	0.0968	0.0358	0.0256	1.260	0.0110
CLIP [24]	0.1501	0.0380	0.0554	2.554	0.0294
SigLIP [25]	0.1449	0.0392	0.0525	2.526	0.0277
T3 [27]	0.0505	0.0250	0.0135	0.671	0.0059
DAv2 [23]	0.0678	0.0275	0.0141	0.701	0.0060
TVL + MAE [†] [20], [29]	0.0523	0.0322	0.0156	0.772	0.0060
Sparsh-X+MAE [†] [20], [30]	0.0561	0.0290	0.0145	0.723	0.0059
VisTacFusion[†]	0.1055	0.0388	0.0138	0.694	0.0058
<i>(b) Tac only</i>					
DAv2 [23]	0.0167	0.0128	0.0474	2.279	0.0116
Sparsh-X [30]	0.0183	0.0138	0.0474	2.268	0.0119
DINOv3 [22]	0.0174	0.0136	0.0663	3.185	0.0162
T3 [27]	0.0177	0.0133	0.2050	9.929	0.0460
SITR [28]	0.0156	0.0118	0.0748	3.626	0.0214
TVL [29]	0.0210	0.0163	0.1057	5.084	0.0247
Sparsh-DINOv2 [21], [26]	0.0189	0.0139	0.1227	5.914	0.0268
Sparsh-MAE [20], [26]	0.0172	0.0130	0.1255	6.030	0.0281
TVL + MAE [†] [20], [29]	0.0231	0.0160	0.0804	3.903	0.0216
Sparsh-X+MAE [†] [20], [30]	0.0199	0.0145	0.0503	2.411	0.0140
VisTacFusion[†]	0.0170	0.0128	0.0488	2.350	0.0137
<i>(c) Both</i>					
VITaL (adapted) [36]	0.0254	0.0182	0.0423	2.034	0.0112
MViTac (adapted) [31]	—	—	0.0215	1.027	0.0072
TVL + MAE [†] [20], [29]	0.0106	0.0103	0.0143	0.720	0.0058
Sparsh-X+MAE [†] [20], [30]	0.0107	0.0101	0.0131	0.655	0.0055
VisTacFusion[†]	0.0066	0.0074	0.0129	0.652	0.0053

Depth MSE is in mm^2 . Rot. L_1 measures errors in $(\cos \theta, \sin \theta)$; Trans. L_1 uses normalized coordinates. Vis/Tac denote UV-off/on input; Both uses the pair. †: weights shared across input configurations within each selected checkpoint; unmarked Vis/Tac rows are dedicated models. Geometry and pose use separate task-selected checkpoints (Sec. V-A). Citations identify source methods; all scores are our evaluations. TVL+MAE and Sparsh-X+MAE are encoder substitutions, and Sparsh-X uses a locally pretrained image encoder.

TABLE II: **Contact geometry and object pose on real validation data.** Bold: lowest error in the paired-input configuration; ties at displayed precision.

the imaging-state change. This result characterizes the tested normal-loading protocol; sliding wear, contamination, and optical drift were not quantified.

Data and geometric references A KUKA LBR Med R820

arm collects approximately 400 contacts for each of 12 objects, recording both image states and end-effector pose at every contact. A reference-object calibration relates object pose to the sensor frame. Pose and object geometry generate the depth reference maps; real normal maps are obtained by finite differences. These are computed geometric targets, not independent measurements of the deformed membrane. For the 12 corresponding geometries, Blender renders visual images, depth maps, and normal maps. Simulated tactile images are synthesized from the rendered depth maps, with random $\pm 15\%$ depth scaling, by a depth-to-tactile generator (U-Net with a PatchGAN discriminator) trained only on real training pairs.

Both image streams are center-cropped to 0.816 of their original dimensions and resized to 224×224 . Simulated visual images receive an additional $1.15\times$ center zoom to align their field of view with real images. Simulated poses are restricted to the per-object real rotation and translation ranges with a margin factor of 1.5. Simulation augmentation applies matched in-plane rotations within $\pm 15^\circ$, together with channel, illumination, and noise perturbations. Real images are not augmented.

Validation and checkpoint selection Every tenth sample within each real session is held out for validation and checkpoint selection. Objects and acquisition sessions are shared between training and validation, and object identity is supplied to the pose queries. All reported scores therefore measure within-session generalization on known objects.

Geometry and pose use separate checkpoints selected by depth MSE and mean angular error, respectively. Fusion models use both for selection; dedicated models use their native mode. Each selected fusion checkpoint is evaluated in all three modes without reselection, ensuring a common checkpoint for each within-task modality comparison.

Geometry and pose metrics Depth and normal MSEs average over all pixels, including non-contact regions. Normal MSE also averages over its three components. Depth MSE is in mm^2 . Rotation L_1 sums the absolute cosine and sine errors per sample; angular error averages the inverse cosine of the clipped predicted-reference dot product, in degrees. Translation L_1 averages both coordinate errors, normalized by half the object’s target size.

Optimization The main fusion runs use real-simulation co-training at a real:sim ratio of 1:1.27. Optimization uses AdamW for 150 epochs with batch size 32, learning rate 2×10^{-4} , weight decay 0.05, cosine decay, and 1,000 warmup steps. We use mixed precision and gradient clipping at 1.0. The input probabilities are $(p_{\text{Vis}}, p_{\text{Tac}}, p_{\text{Both}}) = (0.05, 0.25, 0.70)$, with $p_{\text{inject}} = 0.75$.

Compared estimators All scores in Table II are local evaluations. TVL [29] and Sparsh-X [30] replace the tactile encoder while retaining MAE and our downstream architecture. The Sparsh-X image encoder is locally pretrained on the co-training images and then frozen. VITaL [36] and MViTac [31] use adapted ResNet-18 implementations; MViTac predicts pose only. Dedicated single-modality models attach geometry and pose heads to one frozen encoder. Symmetric MBT [32]

replaces the fusion trunk while retaining the same encoders and heads.

B. Complementary Contributions of Vision and Touch

In Table II, pairing vision and touch reduces VisTacFusion’s depth MSE to 0.0066 mm^2 , corresponding to reductions of 93.7% relative to Vis and 61.2% relative to Tac. Angular error decreases by 72.3%, from 2.350° with Tac to 0.652° with Both, and is also lower than the 0.694° obtained with Vis. Vision therefore provides substantially stronger pose information than touch alone, while both modalities contribute to contact-depth estimation.

Among the paired-input models in Table II, VisTacFusion achieves the lowest error across all five metrics. Compared with Sparsh-X+MAE, it reduces depth MSE by 38.3% and normal MSE by 26.7%, while lowering translation L_1 from 0.0055 to 0.0053. The largest improvements occur in geometry, whereas angular errors remain similar at 0.652° for VisTacFusion and 0.655° for Sparsh-X+MAE.

Dedicated models remain competitive with single input. SITR leads Tac depth and normal accuracy, Sparsh-X leads angular accuracy, and DAV2 leads translation accuracy. With Vis, T3 has the lowest depth, normal, and angular errors, while VisTacFusion has the lowest translation error.

Modality dropout supports geometry and pose inference with either available input using shared parameters. Within VisTacFusion, paired observations give the lowest error on all five metrics, combining this input flexibility with the benefit of complementary measurements.

C. Representation and Training Design

Encoder selection. In Fig. 5(a,b), MAE gives the lowest depth and angular errors for every tactile encoder. SITR+MAE achieves the lowest depth MSE of 0.0067, whereas DAV2+MAE achieves the lowest angular error of 0.618° . We select SITR+MAE to prioritize contact geometry. This sweep compares both pretrained representations and architectures.

Fusion configuration. Using the same encoders and heads, Table III shows that asymmetric fusion reduces paired-input angular error from 0.747° to 0.652° and translation L_1 from 0.0054 to 0.0053. Symmetric MBT achieves lower depth and normal MSEs and the lowest errors across all five Tac metrics. These results characterize a geometry–pose trade-off between the complete fusion configurations: asymmetric fusion uses 16 bottleneck tokens, whereas symmetric MBT uses four.

Computational cost VisTacFusion has 92.5M trainable parameters, whereas symmetric MBT has 83.0M trainable parameters out of 482.9M total. The fusion trunks account for the 9.5M difference in trainable parameters.

Data mixing and modality dropout Figure 6 compares the completed real:sim ratio configurations. At 1:1.27, the main-experiment configuration gives depth MSE 0.0066, angular error 0.652° , and translation L_1 0.0053, compared with 0.0079, 0.720° , and 0.0056 for real-only training. The 1:1.63 configuration also reaches depth MSE 0.0066 at the reported precision; 1:1.27 gives the lowest angular and translation

Input	Model	Geometry		Pose		
		Depth MSE↓	Normal MSE↓	Rot. L_1 ↓	Angle $(^\circ)$ ↓	Trans. L_1 ↓
Vis	Sym. MBT [32]	0.1059	0.0387	0.0168	0.831	0.0069
	VisTacFusion	0.1055	0.0388	0.0138	0.694	0.0058
Tac	Sym. MBT [32]	0.0167	0.0123	0.0452	2.178	0.0120
	VisTacFusion	0.0170	0.0128	0.0488	2.350	0.0137
Both	Sym. MBT [32]	0.0062	0.0071	0.0149	0.747	0.0054
	VisTacFusion	0.0066	0.0074	0.0129	0.652	0.0053

TABLE III: **Fusion configuration comparison with SITR+MAE and identical heads.** Bottleneck tokens: 4 (MBT), 16 (ours). Checkpoints as in Table II; bold: lower error in the paired-input comparison.

errors among the plotted configurations. Real-only and mixed-data runs use 100 and 150 epochs, respectively, with the number of updates also depending on dataset size.

Figure 5(c) varies visual conditioning while fixing the modality probabilities at 0.55 for Both, 0.35 for Tac, and 0.10 for Vis. Depth MSE decreases with more frequent conditioning, whereas angular error varies nonmonotonically. With $p_{\text{inject}} = 0.75$, panel (d) compares different modality mixtures. The selected mixture assigns probabilities of 0.70, 0.25, and 0.05 to Both, Tac, and Vis, respectively, yielding a depth MSE of 0.0065 and an angular error of 0.644° . Paired-only training lowers depth MSE to 0.0058 but increases angular error to 0.675° . These sweeps evaluate paired-input accuracy, while Table II separately assesses robustness to missing modalities.

Qualitative geometry Figure 7 shows errors that full-image averages do not localize. Vis-only predictions miss much of the V-shape and corner contact regions; tactile input recovers their boundaries. Paired input also reduces the irregular rod depth profile visible with Tac alone.

VI. DISCUSSION AND FUTURE WORK

A. Discussion

We presented GelSlim 5.0 and VisTacFusion as a common-view visual–tactile sensing and estimation framework for object pose and contact geometry.

Our results suggest that the value of multimodal sensing lies not only in the distinct strengths of individual modalities, but also in their ability to resolve each other’s ambiguities. Proximity vision provides a global prior for object pose and contact interpretation, whereas tactile sensing supplies a direct local measurement of contact geometry. The further improvement obtained with paired observations indicates cross-modal synergy rather than a simple division of sensing roles, motivating task-specific asymmetric fusion and state-dependent mode selection for dexterous manipulation.

Our real–simulation co-training study further reveals a trade-off between synthetic-data coverage and domain fidelity. Adding simulated observations expands the range of object poses and contact geometries seen during training and improves real-world validation performance over real-only training. However, residual sim–real discrepancies—

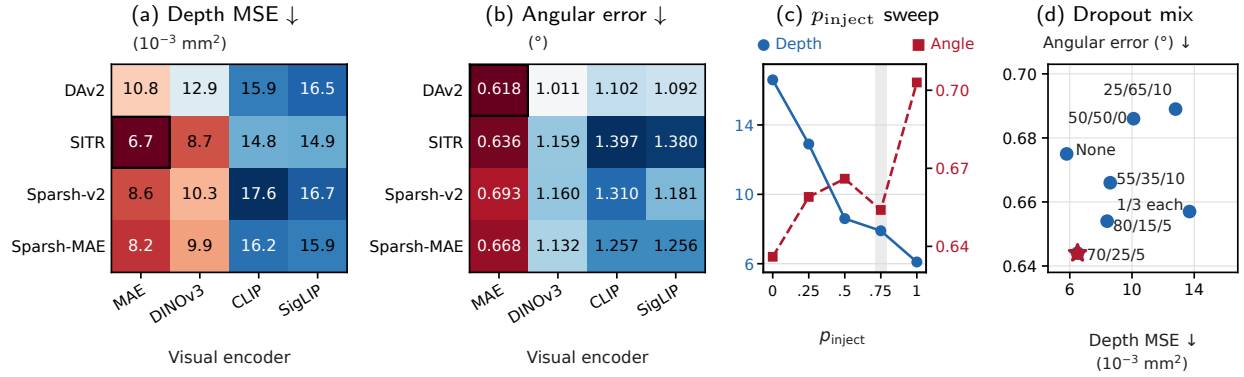


Fig. 5: **Encoder and training ablations with paired input.** (a,b) Depth MSE and angular error for four tactile encoders (rows) and four visual encoders (columns); outlines mark minima. (c) Visual-conditioning sweep with a fixed Both/Tac/Vis mixture of 55/35/10, digitized from the two-seed mean curve. Blue circles show depth MSE (left axis); red squares show angular error (right axis). (d) Modality mixtures at $p_{\text{inject}} = 0.75$; labels give Both/Tac/Vis percentages, and “None” denotes paired-only training. The star marks the selected mixture. Depth is in 10^{-3} mm^2 and angle in degrees. These sweeps are separate from the main-table run.

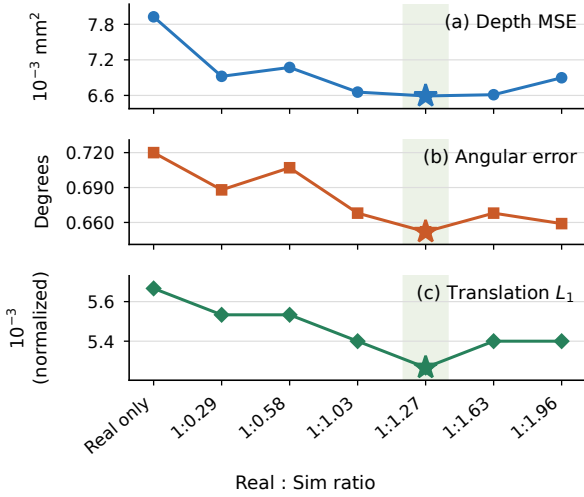


Fig. 6: **Real-simulation data mixing with paired input.** Completed configurations are compared by (a) depth MSE, (b) mean angular error, and (c) normalized translation L_1 . Stars mark the main-experiment ratio of 1:1.27. Depth uses the depth-selected checkpoint; both pose metrics use the angle-selected checkpoint.

including differences in illumination, blur, and background appearance in proximity mode that are difficult to eliminate—can introduce simulation-specific cues. As simulated samples account for a larger fraction of training, these discrepancies may increasingly bias the learned representation away from the real-data distribution, resulting in negative transfer and an intermediate optimal mixing ratio rather than monotonic gains from additional simulation. In our setting, the selected real-to-simulation ratio of 1:1.27 provides the best overall trade-off for pose estimation while retaining competitive geometry accuracy. This optimum is specific to the current simulator, dataset, and training configuration; more broadly, the result suggests that effective sim-real co-training must balance synthetic coverage against domain fidelity.

B. Future Work

Miniaturization and thermal management The current prototype retains the GelSlim 4.0 body; future work will co-design the camera-lens module, illumination, electronics,

and contact stack for a smaller, power-efficient fingertip while preserving field of view and contact sensitivity. We will quantify package size, energy per paired observation, and contact-surface temperature under matched acquisition conditions.

Perception evaluation Future work will evaluate cross-session generalization and validate local contact geometry using contact-region metrics and direct membrane-shape measurements.

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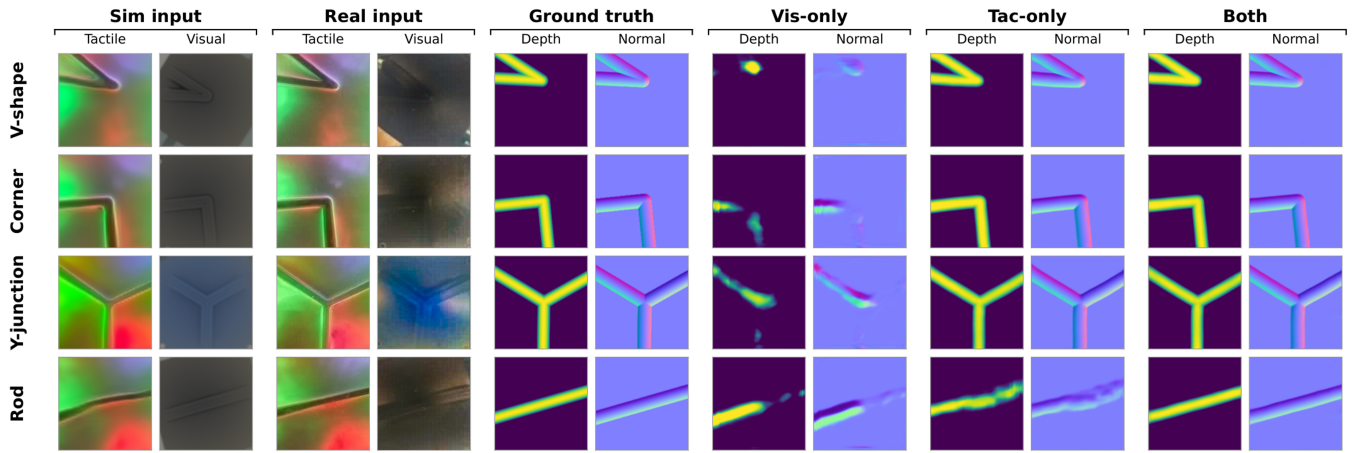


Fig. 7: **Depth and normal predictions on real validation samples.** Sim inputs are shown for comparison: tactile images are generated from each sample’s reference depth, and visual images are rendered at the corresponding pose. Predictions use the real inputs and one depth-selected checkpoint across all three modes. Depth uses a shared color range within each row, and normals are RGB-encoded.

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